

LESSON GUIDE

Technology Plus

Unit 5: Software

3.1.2 – File System Management

Lesson Overview:

In this lesson, students will be introduced to file system management concepts, focusing on how files and folders are organized, named, secured, and described by their attributes. They will examine the importance of proper file and directory structures, permissions and how they control access, and other properties that ensure they are compatible across systems.

Students will:

- Examine how files and folders are structured.
- Explain the purpose of file and folder permissions.
- Identify common naming restrictions across operating systems.
- Describe how file attributes and properties play a role in file management.

Guiding Question: How does file and folder management impact organization, access, and security within a system?

Suggested Grade Levels: 8-12

Technology Needed: Yes

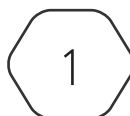
Approximate Time Required: 55-75 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 3.1 – Identify components of an OS
 - File management
 - Folders/directories
 - Permissions
 - Naming Restrictions
 - File attributes and properties

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Lesson Background

Background Information:

Files are containers of data related to the applications and information that users work with day to day in computing. Folders, also called directories, organize files into a hierarchical structure. Most operating systems use file and folder permissions to regulate who can read, write, or execute files. Windows uses NTFS, New Technology File System, permissions such as Read, Write, Modify, and Full Control, while Linux uses rwx notation, or read, write, and execute, for owner, group, and others. Common naming restrictions vary across operating systems. Windows cannot use characters like \ / : * ? " < > | and avoids reserved names like CON, AUX, etc. for specific system files. Linux allows most characters except / and null (\0), but case sensitivity is recognized in Linux while not in Windows. Common file attributes include Read-only, Hidden, System (Windows), or Immutable (Linux). File and folder properties, or metadata, like size, type, created/modified/accessed dates, and owner also contribute to how everything is arranged and structured within a system.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 3.1.2 – File System Management • Lab Slides 3.1.2 – Managing Files in Linux
Materials per Student	• Student Handout 3.1.2 – File System Management • Student Notes 3.1.2 – File System Management • Assessment 3.1.2 – File System Management (Optional)

Lesson Background

Cyber Terms:

Term	Definition
file	the smallest unit of the file system on a device and serves as a container for the data needed for that item, whether it be text, image, video, or software
file attributes	a specific type of metadata that determines how a file or folder is treated by the file and operating systems
file properties	a specific type of metadata that includes the details the user usually sees when viewing a file or folder and assists with user management and organization
folder	used to organize and store files or other folders within a file system on a device, also referred to as a directory
drive	a hardware component of a computing device that stores and retrieves data
metadata	the information and data that describes other data
root directory	the topmost directory in a file hierarchy system

3.1.2 – File System Management

Guiding Question: How does file and folder management impact organization, access, and security within a system?

Lesson Launch (2-3 minutes)

- Start by asking and discussing with students the challenges that might arise if everything on our devices were stored in a single location?
- Reinforce that this means no specific order, random names, all in one location, etc.
 - Encourage further discussion by making sure they understand that this includes system files, and temporary files, most of which they have never interacted with but are required for the device to run properly.

Mastering File Systems (25-35 minutes)

File System Basics

- A **file** is the smallest unit of the file system and serves as a container for the data needed for that item, whether it be text, an image, video, or software. Files may be user files that are interacted with during regular use or system files that work with the device allowing it to function. Files have a wide array of functions and data associated with them that will be discussed later in the lesson.
- A **folder**, sometimes referred to as a directory, is used to organize and store files, or other folders.
- Think of folders as a dresser. That dresser stores multiple drawers, and those drawers store multiple articles of clothing or items that have different purposes.

File Organization

- Operating systems use a hierarchical structure to organize data/information on a device. The start of this structure is called a **drive**, which is where everything is stored. That drive then contains folders with files, or subfolders that have files forming a tree-like structure for organizing everything on the device.
- Note that in this case we are mainly referring to the hard drive of a computer, but there can be external drives, flash drives, and other media that can be referred to as a drive and use the same file structure.
- The image shows the tree-like file structure in Ubuntu.

File Systems Across Operating Systems

- Operating systems all have a **root directory** that serves as a starting point on the drive.
- Windows designates this as C:\ which is commonly called C Drive.
- Linux devices use root and designate it with /.
- MacOS and ChromeOS often hide their root directories with Mac using the Macintosh HD and ChromeOS through developer's mode.
- The root directory is mostly directories containing system files which should not be modified. However, even the user's personal files stem from the root directory.
- Note that students will see this later with the lab, but Windows and Linux can hide files as well but do not have a designated folder to house them nor are they difficult to find.
- Hiding files is crucial to protecting them as they are usually reserved for system files and

determine the functionality of the device.

File Permissions

- **Permissions** define who can do what with a file or folder, which is crucial for security purposes. These were discussed in lesson 1.2.3 - Octal if a refresher is needed.
- Windows uses the NTFS, New Technology File System, which uses the following permissions:
 - **Read** – Users can view files and subfolders.
 - **Write** – Users can add or write to files and subfolders.
 - **Read and Execute** – Users can view and execute, or run, files.
 - **List** – Users can view and list files and subfolders as well as execute files. Often an advanced setting.
 - **Modify** – Users can read, write, and delete files and subfolders.
 - **Full Control** – Users can do anything including changing file permissions.
- Linux and MacOS use the Unix systems of **read**, **write**, and **execute** (rwx). These permissions would be applied to the owner of the files, the owner's group, and other Users. This is why you see three sets of rwx permissions in Linux.
- Note that the d in front of the rwx permissions is there to designate a directory.
- Display the table for students to highlight that the read, write, and execute permissions can be used in any combination to fit the need.
- In a student's account, they usually have Full Control or all rwx permissions to their own files. However, their permissions may be restricted when it comes to modifying existing system files. For example, installing a software program may not be allowed if the user does not have sufficient permissions to make changes to the hard drive and system files/folders needed to install a new program.

File Naming Restrictions

- Naming rules exist just as permissions, to protect certain files and prevent actions or confusion between user files and system files. The rules vary based on operating systems.
- Windows has the following rules:
 - Forbids using the special characters \ / : * ? " < > | to prevent confusion with directories or other characters used for encoding.
 - Names not allowed, but reserved for system files include CON, PRN, AUX, and NUL mainly due to legacy services from MS-DOS, the pre-Windows operating system.
 - Max path length is 260 characters and 255 for a filename in most cases. Note Windows 11 can handle much larger file names, but some legacy services cannot, so it is limited to create uniformity across the system.
 - Case insensitive so file.txt is the same as File.txt
- Linux/macOS :
 - Forbids using the special character / as to prevent confusing it with the root directory.
 - Case sensitive so file.txt is NOT the same as File.txt
 - Allows many special characters, but caution is advised, or extra steps are needed to interact with the file such as enclosing it in quotations.

Metadata Properties and Attributes

- **Metadata** is the information and data that describes other data. Think of metadata as data about data.
- For example, metadata about a file or folder would include the time and date in which that file was created or last modified or could include what user or device accessed it. It varies greatly

based on the type of file being used.

- Attributes and Properties are types of metadata about files and folders that assist with user and system management.
- Attributes are a specific type of metadata that determines how a file or folder is treated by the file and operating systems. Examples include:
 - Immutable – prevents editing or deletion.
 - Hidden from user – prevents accidental editing.
 - Archived – limits interaction and saves space.
 - System – only the OS can interact with it.
- **Properties** are metadata that includes the details the user usually sees when viewing a file or folder and assists with user management and organization. Examples include:
 - File size
 - Creation/modification/access time
 - File type

Lab : Managing Files in Linux (25-35 minutes)

- This lab will use the Ubuntu VM to examine how files and directories are organized and managed, focusing on key properties and attributes.
- Students will interact with a system file called `.bashrc` that determines how the terminal functions within the Ubuntu VM. They will then alter the attributes to prevent modification and further secure the file from accidental edits, deletions, and more.
- Many of the views and interactions that students will examine, especially with the graphical file explorer are found across multiple operating systems with changes involving file location or naming conventions.

Putting It All Together (5-7 minutes)

Discussion Questions

- Why should you use folders and subfolders for long-term file organization?
Student answers will vary. Example responses include it makes it easier to find and work with the files needed. It separates user files from system files, preventing overwriting files that may determine how the device functions.
- How can improper permission settings create security risks on a shared device?
If files and folders are not setup to restrict users who are not the owners or have access to other users' files, they could easily edit or delete them. In some cases, even being able to read the files could be a security risk.
- Describe how file permissions and attributes work together to help manage and protect files.
Permissions limit what users can do, and attributes control how the system handles the file. As an example, there may be a crucial file needed for bootup. The user may not have the permissions to write over the file and the file itself could have the hidden and immutable attributes added to prevent a user from seeing it.